6 **(a)** p.d. = work (done) / charge OR energy transferred from (electrical to other forms) / (unit) charge B1 [1]

(b) **(i)** $R = \rho l / A$ C1

$\rho = 18 \times 10^{-9}$ C1

$R = (18 \times 10^{-9} \times 75) / 2.5 \times 10^{-6} = 0.54\ \Omega$ A1 [3]

(ii) $V = IR$ C1

$R = 38 + (2 \times 0.54)$ C1

$I = 240 / 39.08 = 6.1\ (6.14)$ A A1 [3]

(iii)	$P = I^2R$ or $P = VI$ **and** $V = IR$ or $P = V^2/R$ **and** $V = IR$	C1	
	$= (6.14)^2 \times 2 \times 0.54$	C1	
	$= 41\ (40.7)$ W	A1	[3]
(c)	area of wire is less (1/5) hence resistance greater ($\times$5) OR R is $\propto 1/A$ therefore R is greater	M1	
	p.d. across wires greater so power loss in cables increases	A1	[2]